



IIRSM Course Approval — Full Submission Brief

eLearning Application (No Trainer) | Chainsaw Maintenance & Cross Cutting | Ground Level | Version 1.4

SECTION 1 | Course Identity & Operational Specification

Parameter	Detail
Official Course Title	Chainsaw Maintenance & Cross Cutting (Ground Level Framework)
Course Format	eLearning — Progressive Web Application (PWA) combined with an integrated printed manual; no trainer required
Publisher	Overleaf Publishers Ltd
Course Author / Content Developer	David J Daniel
Copyright	Copyright © 2026 Overleaf Publishers Ltd. All rights reserved. First Edition: July 2026
Platform URL	chainsawcourses.com
Contact Email	info@chainsawcourses.com
Postal Address	69 Newstead Avenue, Orpington, BR6 9RW
NOS Alignment	Independently mapped to UK National Occupational Standards (NOS) for Chainsaw Operations
NPTC Alignment	Assessment criteria independently mapped to City & Guilds NPTC CS30/CS31 unit parameters for theoretical reference and CPD purposes only
CPD Points Awarded	5 Verifiable CPD Points
Guided Learning Hours (GLH)	16 Hours — technical text, annotated diagrams, and seven sequential video modules
Directed Online Assessment	1.5 Hours — 45-question randomised multiple-choice examination
Independent Self-Study	1.5 Hours — risk evaluation practice, glossary review, and field tool exercises
Total Qualification Time (TQT)	19 Hours total
Minimum Pass Threshold	80% or higher on the summative examination — unlimited resit attempts permitted
Target Learner Profile	Commercial chainsaw operators, forestry workers, arborists, estate and grounds teams, and landscape professionals
Course Version	Version 1.4 — submitted for initial IIRSM eLearning Course Approval (1 course)
Approval Category	New to IIRSM Course Approval — eLearning (no trainer)

SECTION 2 | Developer & Author Credentials

IIRSM requires full details of who developed the course and their level of knowledge and skills to do so. The following provides a complete account of the authorship and development process for this eLearning programme.

- **Course Author:** David J Daniel — specialist course author with comprehensive practical and theoretical knowledge of chainsaw safety, maintenance, and ground-level cross-cutting operations. Author of the companion 21,000-word technical manual (Chainsaw Manual v1.4) which forms the primary learning text for this programme.
- **Publisher:** Overleaf Publishers Ltd — registered UK publisher and copyright holder. Responsible for course design, platform development, content quality assurance, and ongoing maintenance of all course materials.
- **Content Development Basis:** Course content was developed with direct reference to: HSE chainsaw guidance documentation; PUWER 1998; COSHH 2002; Health and Safety at Work etc. Act 1974; City & Guilds NPTC CS30 and CS31 unit standards (used for alignment mapping only — no formal affiliation); Forestry Commission and APHA biosecurity guidance including active Statutory Plant Health Notices; Oregon, Stihl, and Husqvarna manufacturer chain specification data; and UK arboricultural and forestry industry codes of practice.
- **Technical Accuracy:** The chainsaw chain identification appendix has been verified against current manufacturer published specifications. All legal references cite current UK primary and secondary legislation applicable at the time of publication (July 2026). The manual and app carry a clear disclaimer of non-affiliation with any awarding body.
- **Copyright and Intellectual Property:** The original prose, instructional formatting, rephrased revision questions, diagrams, and structured layout of both the manual and digital platform constitute original intellectual property of Overleaf Publishers Ltd. No part may be reproduced without prior written authorisation.
- **Ongoing Development Commitment:** Course content will be reviewed and updated in line with: legislative changes, new APHA/Forestry Commission biosecurity notices, delegate feedback patterns, and manufacturer specification updates. Version history is tracked and the platform is updated accordingly.



SECTION 3 | Digital eLearning Delivery Platform & Quality Gateways

The chainsawcourses.com platform is a Progressive Web Application (PWA) accessible on any device with a modern browser. IIRSM assessors are invited to request direct platform access for evaluation. The following integrity controls are implemented across all seven sequential training modules:

- **Sequential Module Locking:** Each of the 7 modules unlocks only after the preceding module video is fully watched and the associated knowledge quiz is passed at 80% or above. Students cannot skip ahead or access content out of sequence at any point.
- **Anti-Scrubbing Enforcement:** Video timeline scrubbing is disabled for all first-time viewings. Students must engage with the complete video duration to satisfy GLH requirements and unlock the post-module quiz. Every seek attempt is logged to the database with timestamp and attempt count for audit purposes.
- **Device-Locked Single-Use Activation:** Each activation code is purchased individually and bonds atomically to the first device UUID encountered on first login. The code cannot be used on a second device unless an administrator explicitly resets the device bond, preventing credential sharing and ensuring one learner per licence.
- **Dynamic Personalised Watermarking:** A floating watermark displaying the student's registered full name and email address overlays all video content throughout the viewing session and repositions automatically every 60 seconds. This provides unambiguous attribution of all viewed content and acts as a strong deterrent against screen-recording for redistribution.
- **Heartbeat Progress Logging:** Video playback positions are written to the secure database at 30-second intervals, producing a verifiable timestamped engagement trail per student per module. This audit trail directly supports GLH verification requirements.
- **Legally-Binding Digital Waiver:** Before accessing any course content, all students must read and individually tick seven contractual clauses covering: Educational Intent Only, No Certification Conferred, Regulatory Compliance, Personal Protection and Absolute Sobriety, Exclusion of Liability, Disclaimer of Warranties, and Prohibition of Lone Working with Mandatory Emergency Supervision. A legally-binding digital signature is captured via touch or mouse and the completed waiver is rendered as a permanently stored PDF — accessible to administrators at any time.
- **Student Progress Dashboard:** The admin dashboard provides real-time visibility of all enrolled students, showing module completion status, quiz attempt history with scores, video engagement events, inspection records, risk assessments, signed waiver PDFs, and device bond status.

SECTION 4 | Evidence of Quality Assurance & Continuous Improvement

IIRSM requires evidence of quality assurance processes, including how delegate feedback is collected and fed back into course development. The following mechanisms are built into the platform and operated by Overleaf Publishers Ltd:

- **Per-Module Delegate Feedback:** A structured feedback form is presented to every student on completion of each module, collecting a 1-to-5 star rating and an optional written comment. All responses are stored in the database with the student name, module title, rating value, written comment, and submission timestamp.
- **Feedback Retention Policy:** All feedback records are stored indefinitely in a persistent cloud-hosted PostgreSQL database — exceeding the IIRSM minimum 12-month retention requirement. Overleaf Publishers Ltd confirms that all delegate feedback collected since platform launch will be retained and made available to IIRSM upon request for up to and beyond the 18-month approval period.
- **Admin Feedback Dashboard:** The administrator feedback panel provides real-time aggregate star ratings across all modules, a searchable feedback log filterable by module, student name, and comment content, and the calculated average rating across all submissions. This enables immediate identification of low-scoring modules for content review.
- **Quiz Performance Analytics:** All quiz attempts are logged with score, pass or fail outcome, and timestamp. Persistently low pass rates on specific modules or question clusters are reviewed by the content team and used to guide revisions to explanatory text, video content, or question phrasing.
- **Video Engagement Analytics:** Seek attempt counts, completion status, and watch timestamps are logged per student per module. Modules with elevated seek attempt rates or low completion rates are reviewed for pacing, length, or content clarity issues.
- **Summative Examination Monitoring:** All 45-question summative exam attempts are logged with individual question-level answer records, enabling statistical analysis of which questions are most frequently answered incorrectly. This data directly informs future question bank revisions and content updates.
- **Version Control & Change Log:** Course content is version-controlled. The current submission is Version 1.4 (July 2026). Any future updates to course content, assessment questions, or platform features will increment the version number and be logged with a description of changes made and the rationale.
- **IIRSM Membership Promotion:** The platform and associated marketing materials actively promote the benefits of IIRSM membership to all enrolled delegates. The 15% IIRSM membership discount for attending delegates will be communicated through the post-completion certificate page and email confirmation.

**SECTION 5 | IIRSM Risk Management & Leadership Competency Framework Alignment**

The following table maps each element of the IIRSM Risk Management and Leadership Competency Framework to the corresponding course Learning Outcomes and Assessment Criteria. As a ground-level operative training programme, all competencies are addressed at the OPERATIONAL attainment level — Knowledge and understanding, with some application.

IIRSM Competency Domain	Attainment Level	Course Learning Outcome Mapping
Legal & Regulatory Compliance	Operational	LO1: AC 1.1 (HSWA 1974 employer/employee duties), AC 1.2 (PUWER 1998 equipment obligations), AC 1.3 (COSHH 2002 substance risk controls), AC 1.4 (CE/UKCA PPE standards and class markings)
Hazard Identification & Risk Evaluation	Operational	LO2: AC 2.1 (5-step site risk assessment), LO6: AC 6.1 (tension and compression force analysis in timber), AC 6.5 (windblown windfalls — stored energy, stem roll, canopy release hazards)
Risk Control Implementation	Operational	LO1: AC 1.4 (PPE selection and use), LO4: AC 4.1-4.4 (equipment maintenance to prevent mechanical failure), LO5: AC 5.1-5.2 (pre-start and pre-use inspection protocols as mandatory controls)
Emergency Planning & Response	Operational	LO2: AC 2.2 (emergency extraction plan with OS National Grid reference, What3Words, postal address, trauma kit location, and access route for emergency services)
Environmental & Biosecurity Risk	Operational	LO2: AC 2.3 (biosecurity cleaning protocols to prevent spread of Ash Dieback, Sudden Oak Death, Sweet Chestnut Blight, Spruce Bark Beetle per APHA SPHNS)
Technical Equipment Safety	Operational	LO3: AC 3.1 (battery vs IC platform risk comparison including thermal runaway), AC 3.2 (10 integrated chainsaw safety components and their mechanical functions)
Safe Operational Procedures	Operational	LO5: AC 5.1-5.2 (startup and 4-point pre-use dynamic check), LO6: AC 6.2-6.4 (cutting technique, kickback prevention, plunge bore, tension release sequences)
Substance & Exposure Risk Control	Operational	LO1: AC 1.3 (COSHH assessment for two-stroke fuels, alkylate oils, chain lubricants, lithium-ion cells, and toxic arboreal species dusts)
Knowledge of Professional Standards	Operational	LO1: AC 1.1-1.4 (statutory legislation), LO4: AC 4.2-4.4 (equipment maintenance standards), LO6: AC 6.3-6.5 (industry cutting standards and escape route protocols)

SECTION 6 | Programme Learning Outcomes & Assessment Criteria

Six Learning Outcomes (LO) across three teaching units, each with mapped Assessment Criteria (AC), independently aligned to UK NOS and City & Guilds NPTC parameters. All LOs are assessed in the 45-question summative examination except where explicitly noted.

UNIT 1 | Occupational Standards, Health & Safety, and Site Risk Evaluation**LO1: Understand the statutory legal framework and personal safety obligations governing chainsaw ground operations.**

AC 1.1: Identify the primary duties of care placed on employers and employees under the Health and Safety at Work etc. Act 1974 (HSWA 1974), including the duty to provide safe plant, safe systems of work, adequate information and training, and a safe working environment, and recognise equivalent frameworks applicable to international operators.

AC 1.2: Explain the operational and maintenance obligations mandated by the Provision and Use of Work Equipment Regulations 1998 (PUWER), covering regular inspection schedules, operator competency requirements, suitability of equipment for the task, and mandatory safety feature standards for chainsaws in commercial and professional operations.

AC 1.3: Summarise the substance risk assessment and exposure control requirements under COSHH 2002 as applied to two-stroke petrol-oil fuel blends and alkylate alternatives, bar and chain lubricating oils, lithium-ion battery cell handling and disposal, and inhalation and dermal exposure risks from toxic arboreal species including oak processionary moth frass, yew, and certain hardwood dusts.

AC 1.4: Detail the correct CE/UKCA standard class markings required for personal protective equipment including safety helmets with integral visor and ear defenders, cut-resistant chainsaw gloves, chainsaw safety footwear, and Type A versus Type C protective chainsaw trousers — including manufacturer shelf-life limitations and mandatory pre-use inspection and replacement obligations.

LO2: Evaluate environmental site hazards, formulate a structured risk assessment, and establish robust emergency response protocols.

AC 2.1: Execute a 5-step site-specific risk assessment identifying and recording: ground hazards including uneven terrain, root plates, and loose ground; slope and gradient dynamics; proximity of pedestrians and public access; overhead targets including power lines, dead branches, and adjacent trees; and structural vulnerabilities within the planned working area.

AC 2.2: Formulate a complete emergency communication and extraction plan containing: an accurate OS National Grid reference, postal address, What3Words location identifier, site access route, designated meeting point for emergency services, and the confirmed physical location of the dedicated chainsaw trauma first-aid kit containing wound dressings and a tourniquet.



AC 2.3: Identify the mandatory biosecurity tool-cleaning controls required to interrupt transmission of regulated invasive arboreal pathogens between sites, including *Hymenoscyphus fraxineus* (Ash Dieback), *Phytophthora ramorum* (Sudden Oak Death), *Cryphonectria parasitica* (Sweet Chestnut Blight), and *Ips typographus* (Spruce Bark Beetle), in line with active APHA Statutory Plant Health Notices (SPHNs).

UNIT 2 | Power Unit Architecture, Mechanical Integrity, and Component Maintenance

LO3: Analyse the design attributes, operational differences, and integrated safety feature architecture of internal combustion and battery-powered chainsaw platforms.

AC 3.1: Compare the operational risks, charging hazard profile including lithium-ion thermal runaway triggered by impact deformation, moisture ingress, or incorrect charger use, battery state management requirements, and commercial performance limitations of battery-powered units against internal combustion petrol platforms across forestry and arboricultural operational contexts.

AC 3.2: Map and explain the mechanical function of the 10 core integrated safety components spanning the front, centre, and rear architecture of a standard chainsaw: Front Hand Guard and Inertia Chain Brake, Exhaust Muffler with Spark Arrestor, Low-Kickback Bar Nose Loop Profile, Chain Catcher, Anti-Vibration Isolation Mounts, On/Off Kill Switch, Throttle Lockout Interlock Trigger, Mandatory Safety Decals, Right-Hand Rear Guard Flare, and Bar Scabbard.

LO4: Diagnose, service, and maintain the full mechanical integrity of a commercial chainsaw cutting assembly.

AC 4.1: Detail the correct daily air filtration housing cleaning procedure and interpret spark plug electrode colour indicators to assess in-cylinder combustion health: light-tan or coffee-brown indicates a correct mixture; grey or white indicates a lean condition or overheating; sooty black indicates a rich mixture, oil fouling, or air filter blockage requiring immediate rectification.

AC 4.2: Explain safe carburettor performance adjustment protocols, identifying the function and factory limit constraints of the Idle speed (LA/T), Low-speed (L), and High-speed (H) needle screws, and explain why unskilled over-adjustment risks catastrophic lean seizure at operating temperature or uncontrolled RPM runaway.

AC 4.3: Differentiate between Rim and Spur (Star) drive sprockets, identify correct wear replacement thresholds at 0.5mm groove depth, and diagnose guidebar rail wear conditions including lateral burr formations, rail channel splaying, and thermal bluing indicating chronic overheating from inadequate bar oil flow.

AC 4.4: Identify chain pitch, gauge, and drive-link count specifications; distinguish between Full-Chisel and Semi-Chisel tooth geometries and their respective cutting performance, durability, and dulling characteristics; and calculate correct sharpening profiles applying top-plate filing angles in the 25-to-35-degree range, side-plate depth gauge clearances using a raker tool, and an 85-degree wave hook parameter on the side plate.

UNIT 3 | System Startups, Operational Proving, and Ground-Level Timber Processing

LO5: Implement safe pre-start inspection protocols and machine startup methodologies in accordance with manufacturer guidance and industry best practice.

AC 5.1: Execute a structured pre-start visual inspection covering: chain tension correctness, chain sharpness and tooth integrity, chain brake engagement and release function, guidebar straightness and groove cleanliness, chain oiler function with reservoir level, correct fuel mixture or full battery charge, air filter condition and seating, handguard and chain catcher integrity, and exhaust muffler and spark arrestor security. Differentiate between the cold-start floor-anchor technique and the upright knee-clamp warm-start method.

AC 5.2: Perform the mandatory 4-point pre-use dynamic check: confirm chain brake engages and chain stops when bar guard is pushed forward; verify oil spray dispersion to bar and chain using a spray-line surface test; confirm chain does not creep at engine tick-over idle; confirm the on/off kill switch cuts the engine immediately when operated.

LO6: Apply mechanical principles to safely identify, manage, and resolve tension and compression forces during ground-level timber cross-cutting operations.

AC 6.1: Analyse a downed log or supported stem to determine where compression (closing) and tension (opening) forces reside within the wood fibre structure, and select the correct cutting entry face and sequence to eliminate guidebar pinching, uncontrolled timber movement, and splitting hazards.

AC 6.2: Explain the physical difference between a pulling chain cut on the bottom bar face and a pushing chain cut on the top bar face, and apply this understanding to prevent guidebar jams, reactive machine movement, and loss of operator control during cross-cutting operations.

AC 6.3: Define the upper nose quadrant kickback danger zone of the guidebar, explain the rotational inertia mechanism that triggers reactive kickback when the bar nose contacts wood or is pinched under load, and describe how to execute a controlled plunge bore entry cut safely by engaging with the lower bar face first and pivoting through while avoiding nose contact with the kerf walls.

AC 6.4: Sequence and execute safe cross-cutting workflows for: standard ground-supported logs; oversized timber requiring multiple-pass reduction; advanced step cuts for supported log ends; box splits for timber under extreme compression; snedding branch removal; and extreme tension scenarios using the Toast Rack graduated reduction or sink-cut method to incrementally release stored energy before final severance.

AC 6.5: Identify the severe hazard vectors when processing windblown windfalls — stored energy in bent or partially uprooted root plates, unpredictable stem roll on slope, entangled canopy tension release, and rapid ground movement — and establish a pre-planned escape route at 45 degrees to the rear before committing any cuts to the root plate connection or main stem.

SECTION 7 | Integrated Field Tools & Supplementary Resources (Non-Assessed)



In addition to the seven sequential training modules, activated students have access to the following standalone professional tools. These are non-assessed practical aids designed to extend theoretical learning into real-world operational contexts and reinforce the risk management behaviours aligned to the IIRSM Competency Framework:

- **Pre-Start & Pre-Use Inspection Checklist:** 12-item pre-start check and 4-item pre-use dynamic check logged to the database with saw identifier and submission timestamp. Failed items are prominently flagged. All records are permanently stored and reviewable by administrators — directly reinforcing LO5 AC 5.1 and AC 5.2.
- **Dynamic Risk Assessment Tool:** GPS-derived site location auto-populating an OS National Grid reference, address, and What3Words identifier. Operators record task description, site description, access route, meeting point, and first-aid kit location. An editable 11-item hazard matrix covers kickback, entanglement, manual handling, trips and falls, trapped guidebars, rolling timber, HAVS, fuel fire risk, weather and visibility, bystander exclusion zones, and lone working — each scored by Likelihood x Severity (1-5) generating a Low, Medium, or High risk rating. Directly reinforces LO2 AC 2.1 and AC 2.2.
- **Biosecurity & Hazard Zone Map:** Interactive Leaflet map with toggleable illustrative hazard overlays: Direct Operator Health Hazards (Oak Processionary Moth, Brown-tail Moth) and Statutory Containment Zones (Spruce Bark Beetle, Ash Dieback, Sudden Oak Death, Sweet Chestnut Blight, Western Hemlock Dieback). Students are directed to verify active zone boundaries against live Forestry Commission and APHA notices before commencing operations. Directly reinforces LO2 AC 2.3.
- **Chain Identification Chart:** Searchable reference tool reproducing the manual appendix (page 137). Select brand (Oregon, Stihl, or Husqvarna) and enter a drive-link count to retrieve chain pitch, gauge, correct file diameter, and top-plate filing angle, plus the full reference table, pitch-to-engine-power guide, and chain designation letter code glossary. Reinforces LO4 AC 4.4.
- **Timber Species Reference Guide:** Reference covering UK native and non-native tree species relevant to chainsaw operations — cutting difficulty, firewood quality grade, seasoning duration, spit risk rating, commercial and craft uses, active pest and disease threats, and operator safety warnings for toxic species and hazardous wood dusts.
- **Cross-Cutting Scenario Simulator:** Interactive SVG-based tension and compression decision trainer presenting three realistic log loading scenarios: sag (downward bow), hog (upward bow or crown loading), and sidebend (lateral lean). Students select a cut position; correct entries produce audio success feedback while incorrect entries trigger a pinch-error sound with guidance — directly reinforcing LO6 AC 6.1 in a low-risk digital environment.
- **Industry News Feed:** Curated forestry and arboriculture industry news sourced from approved RSS feeds with administrator review before publication. Students may opt in to web push notification subscriptions to receive breaking news alerts including legislation changes, APHA biosecurity notices, and equipment recalls — supporting ongoing professional awareness in line with the IIRSM CPD ethos.

SECTION 8 | Scope Limitation & Theory-Only Boundary

- **Theory-Only CPD Provision:** This programme provides theoretical knowledge verification and structured CPD hours only. It does not provide delegates with the legal authorisation or certification required to operate a chainsaw commercially or professionally. This is explicitly confirmed in the 7-clause digital liability waiver signed by every student before accessing any course content.
- **No Practical Competency Certification:** The mandatory statutory requirement for face-to-face practical training and independent field assessment — City & Guilds NPTC CS30 (Chainsaw Maintenance and Crosscutting) or CS31 (Felling Small Trees) — remains entirely separate from and unaffected by completion of this qualification. This programme functions as a CPD study aid and theoretical preparatory resource only.
- **IIRSM Approval Scope:** This application is submitted strictly in the category of CPD theoretical eLearning and does not claim to confer any practical qualification, formal certification, or legal operating authority. The programme is therefore considered eligible under IIRSM's approval criteria which explicitly accommodates risk-related CPD courses that do not grant formal qualifications or legal authorisations.

SECTION 9 | Legal Framework — Terms & Conditions, Data Protection & Complaints

The following summarises the legal and compliance framework operated by Overleaf Publishers Ltd for this course. Full policy documents are published at chainsawcourses.com and are available to IIRSM upon request.

- **Terms & Conditions and Liability Waiver:** All students must accept a 7-clause legally-binding liability waiver before accessing any course content. Clauses cover: (1) Educational Intent Only; (2) No Certification Conferred; (3) Regulatory Compliance — operator retains sole responsibility for compliance with HSWA, PUWER, and applicable codes of practice; (4) Personal Protection and Absolute Sobriety — mandatory PPE use and prohibition of operation while fatigued or impaired; (5) Exclusion of Liability — Overleaf Publishers Ltd disclaims liability for injuries, losses, or misapplication of content to the maximum extent permitted by law; (6) Disclaimer of Warranties; (7) Prohibition of Lone Working — student explicitly acknowledges they must never operate a chainsaw alone and that a second competent person with a trauma first-aid kit must be present at all times. Digital signature is captured and timestamped.
- **Refund & Cancellation Policy:** Activation codes are single-use licences sold through the chainsawcourses.com checkout. Once an activation code has been bonded to a device and course access has commenced, no refund is available as digital content has been delivered. Pre-access cancellation requests must be submitted in writing to info@chainsawcourses.com within 14 days of purchase. Full policy is available at chainsawcourses.com.



- **Data Protection Policy (UK GDPR):** Overleaf Publishers Ltd is the data controller. Lawful basis: contract performance (Art. 6(1)(b) UK GDPR). Data collected includes: full name and email address, activation code, device UUID, video engagement timestamps, quiz attempt records, digital waiver signature and timestamp, inspection records, and risk assessment records. Data is stored in a secure cloud-hosted PostgreSQL database. No personal data is shared with third parties. Students may exercise UK GDPR rights — access, rectification, erasure — by contacting info@chainsawcourses.com. Full privacy policy at chainsawcourses.com/privacy.
- **Complaints Procedure:** All complaints regarding course content accuracy, platform access, or assessment outcomes should be submitted in writing to info@chainsawcourses.com. Complaints will be acknowledged within 3 working days and a formal response issued within 14 working days. Where a complaint relates to content accuracy, the material will be reviewed and, if an error is confirmed, corrected and the version number incremented. Records of all complaints and actions taken are retained for a minimum of 3 years.

SECTION 10 | Advanced Workshop Enrichment Content (Non-Assessed)

The following content is included in both the printed manual and companion app strictly for professional enrichment and advanced field awareness. Each item carries the Advanced Workshop Extension Module designation and is entirely distinct from the assessed LO/AC framework — it will not appear in the summative examination and is not mapped to any assessed criterion.

1. **Two-Stroke Combustion Cycle & Fuel Calculation:** The Suck-Squeeze-Bang-Blow four-event combustion sequence and standard 50:1 fuel-to-two-stroke-oil volume mixing calculations are provided for background mechanical understanding only. This content is explicitly excluded from the assessed syllabus.
2. **Oil Pump & Clutch Assembly Overhaul:** Extraction of the centrifugal clutch shoe assembly to expose the underlying automated bar oil pump worm drive, metering adjuster, and internal pickup tube — advanced preventative maintenance beyond standard daily servicing scope.
3. **Advanced Windblown Tree Processing:** Managing complex energy vectors in tangled root-plate windfalls, multi-directional stem loading analysis, and safe sequenced canopy clearance. Content provided for professional awareness; practical application requires face-to-face specialist training.
4. **Winch Rigging & Multi-Redirect Pulley Systems:** Configuring heavy-capacity winch systems with multi-redirect pulley blocks and anchor rigging to stabilise large root plates and control stem fall direction before committing cuts. Professional arboricultural rigging technique — for awareness only.

Submitted For IIRSM Course Approval By

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Authorised Signatory — Overleaf Publishers Ltd